Structure-Aware Keyphrase Extraction with Positional, Section, and Semantic Relevance Weighting

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Abstract

Keyphrase extraction from scientific documents is a fundamental task in natural language processing. While TF-IDF provides a strong unsupervised baseline, it ignores the structural information inherent in academic papers. We propose SAKE (Structure-Aware Keyphrase Extraction), a multiplicative scoring framework that augments TF-IDF with three additional signals: positional weighting (phrases appearing earlier score higher), section-aware weighting (title and abstract phrases are boosted), and semantic relevance via Latent Dirichlet Allocation (LDA). Through an ablation study on the Krapivin benchmark (2,304 documents), we find that section-aware weighting is the most impactful component, improving F1@5 by 28.7% relative to TF-IDF with positional features alone. Positional weighting contributes a further 12.0% improvement. However, adding LDA-based semantic relevance yields no additional gain, suggesting that lightweight topic models are redundant in the presence of explicit structural signals. Our best configuration (TF-IDF + Position + Section) achieves an F1@5 of 0.1539, and aggressive section weights (title 8×, abstract 4×) push this to 0.1668, a 56.2% relative improvement over the TF-IDF baseline.

Keywords: Keyphrase Extraction, TF-IDF, Document Structure, Positional Weighting, LDA, Ablation Study

1 Introduction

Keyphrase extraction is a core task in natural language processing that supports downstream applications such as information retrieval, summarization, topic indexing, and scientific document understanding. TF-IDF serves as a strong, interpretable baseline for this task because it is efficient and easy to implement. However, TF-IDF treats a document as a bag of words and fails to model structural information that is critical in long, structured documents like research papers [1].

In scientific articles, candidate phrases appearing in the title, abstract, or early sections are typically more informative than those found only in later body text. Prior work has demonstrated that positional information and semantic relevance can improve unsupervised keyphrase extraction quality [1–5].

This project investigates whether incorporating positional and structural signals into a TF-IDF-based framework can improve keyphrase extraction performance while maintaining interpretability, and whether a lightweight notion of topic relevance via Latent Dirichlet Allocation (LDA) [6] can further enhance keyword ranking.

2 Experimental Plan

2.1 Working Hypotheses

We formulate three hypotheses that our experiments aim to test:

H1 (Structural Salience). Candidate phrases appearing in salient sections (title, abstract) are more informative than those in regular body text. Assigning higher structural weights to title and abstract phrases should improve precision in identifying primary keyphrases.

H2 (Positional Salience). Phrases located earlier in the document carry higher topical weight. Applying positional weighting based on first-occurrence position should improve recall and F1.

H3 (Semantic Relevance). A lightweight topic model (LDA) can capture topical fit beyond raw frequency, further improving ranking quality when combined with positional and section-aware features.

2.2 Dataset

We use the Krapivin Benchmark [7], a collection of 2,304 full-text computer science research papers with author-assigned extractive keyphrases as ground truth. Of these, 2,283 documents have non-empty ground truth and are used for evaluation. No training split is needed since our approach is entirely unsupervised. The dataset is loaded from HuggingFace (`midas/krapivin`, `raw` config).

Preprocessing.

The preprocessing pipeline (implemented in `ATP.ipynb`) consists of:

1. **Document Parsing**: Tokens are segmented into title, abstract, and body sections using the `--` delimiter and section identifiers (T, A, B).
2. **Text Normalization**: Regex-based cleaning removes LaTeX equations, citations, emails, URLs, section numbers, and special characters. Text is lowercased.
3. **Candidate Phrase Extraction**: spaCy (`en_core_web_sm`) extracts noun phrases using POS patterns: $(\text{Adj})^*(\text{Noun})^+$. Stopwords, punctuation-only strings, and excessively short/long phrases are removed.
4. **Output**: A preprocessed pickle file (`preprocessed_krapivin.pkl`) containing per-document: `doc_id`, `title_phrases`, `abstract_phrases`, `body_phrases`, `all_phrases` (concatenation preserving order), and `ground_truth`.

2.3 Experimental Design

Table 1 lists the fixed parameters and Table 2 lists the free parameters explored in the sensitivity analysis.

Table 1 Fixed parameters

Parameter	Value	Rationale
POS pattern	$(\text{Adj})^*(\text{Noun})^+$	Standard practice
spaCy model	`en_core_web_sm`	Lightweight, sufficient for POS
IDF smoothing	$\log\left(\frac{N}{1+\text{df}}\right)$	Standard TF-IDF
Evaluation cutoffs K	5, 10, 15	Typical keyphrase list lengths
Stemmer	Porter Stemmer	Standard for evaluation
LDA topics	20	Granularity vs. generalization
LDA passes	10	Sufficient for convergence
LDA random seed	42	Reproducibility
Position weight	$\frac{1}{1+\text{pos}/\text{total}}$	Smooth decay $[0.5, 1.0]$
Topic weight shift	$0.5 + \cos(\mathbf{d}, \mathbf{p})$	Modulating range $[0.5, 1.5]$

Table 2 Free parameters (sensitivity analysis)

Parameter	Tested Values	Purpose
Title weight α_T	2.0, 3.0, 5.0, 8.0	Sensitivity analysis
Abstract weight α_A	1.5, 2.0, 3.0, 4.0	Sensitivity analysis
Body weight α_B	1.0 (fixed)	Reference level

2.4 Experimental Setup

Combined Scoring Model.

For each candidate phrase t in document d, the final score is computed as:

$$\text{Score}(t) = \text{TF-IDF}(t) \cdot W_{\text{position}}(t) \cdot W_{\text{structure}}(t) \cdot W_{\text{topic}}(t), \tag{1}$$

where

$$W_{\text{position}}(t) = \frac{1}{1 + \frac{\text{first_pos}(t)}{\text{total_phrases}}}, \tag{2}$$

$$W_{\text{structure}}(t) = \max\big(\alpha_B,\ [\,t \in \text{Abstract}\,] \cdot \alpha_A,\ [\,t \in \text{Title}\,] \cdot \alpha_T\big), \tag{3}$$

$$W_{\text{topic}}(t) = 0.5 + \cos(\mathbf{d}_{\text{LDA}},\ \mathbf{t}_{\text{LDA}}). \tag{4}$$

Ablation Study Design.

We isolate the contribution of each feature through incremental addition:

1. **TF-IDF only** — bag-of-words frequency baseline.
2. **TF-IDF + Position** — adds Eq. (2).
3. **TF-IDF + Position + Section** — adds Eq. (3).
4. **TF-IDF + Position + Section + Semantic (Full SAKE)** — adds Eq. (4).

Evaluation Metrics.

We use Precision@K, Recall@K, and F1@K, with predictions matched to gold keyphrases after Porter stemming (standard practice [1]). Results are macro-averaged across all 2,283 documents with ground truth.

3 Results

3.1 Ablation Study

Table 3 presents the main ablation results.

Table 3 Ablation study — Precision, Recall, and F1 at $K = 5, 10, 15$. Best values per metric in bold.

	$K = 5$			$K = 10$			$K = 15$		
Model	P	R	F1	P	R	F1	P	R	F1
TF-IDF	.0979	.1334	.1068	.0789	.2145	.1104	.0664	.2700	.1027
+ Position	.1093	.1496	.1196	.0859	.2340	.1201	.0722	.2957	.1117
+ Pos + Sec	**.1400**	**.1938**	**.1539**	**.1050**	**.2899**	**.1470**	**.0845**	**.3473**	**.1307**
Full SAKE	.1391	.1938	.1534	.1031	.2857	.1446	.0835	.3461	.1293

Positional weighting (H2: Supported).

Adding position weights improves F1@5 from 0.1068 to 0.1196 (+12.0% relative), confirming that phrases appearing earlier in academic papers are more likely to be keyphrases.

Section-aware weighting (H1: Strongly Supported).

This is the most impactful component. F1@5 jumps from 0.1196 to 0.1539 (+28.7% relative) when section weights are added. Title and abstract phrases receive 3× and 2× boosts respectively, which dramatically improves both precision and recall. This aligns with the hypothesis that structural salience in scientific documents provides the strongest signal for keyphrase identification.

Semantic relevance via LDA (H3: Not Supported).

Adding the LDA topic weight slightly *decreases* performance (F1@5: 0.1539 → 0.1534; F1@10: 0.1470 → 0.1446). The semantic signal from a 20-topic LDA model appears

largely redundant with position and section features, or the topic granularity may be insufficient to distinguish keyphrases from related but non-key phrases within the same topic cluster.

3.2 Hyperparameter Sensitivity

Table 4 shows F1 for different section weight configurations using the TF-IDF + Position + Section model.

Table 4 F1 at different section weight configurations

Configuration	F1@5	F1@10	F1@15
Low: α_T=2.0, α_A=1.5	.1446	.1367	.1252
Mid: α_T=3.0, α_A=2.0	.1539	.1470	.1307
High: α_T=5.0, α_A=3.0	.1625	.1535	.1354
Very High: α_T=8.0, α_A=4.0	**.1668**	**.1580**	**.1368**

Performance increases monotonically as section weights grow larger, suggesting the model benefits from aggressively boosting title and abstract phrases. The "Very High" configuration achieves F1@5 of 0.1668, a 56.2% relative improvement over the TF-IDF baseline. This strong sensitivity to section weights underscores that document structure is the dominant feature for keyphrase extraction in scientific papers.

3.3 Error Analysis

Table 5 breaks down missed keyphrases by n-gram type for the Full SAKE model at K=15.

Table 5 Missed keyphrases by n-gram type (Full SAKE, K=15)

N-gram Type	Missed Count	Percentage
Bigram	3,534	61.8%
Unigram	1,152	20.2%
3+ gram	1,032	18.1%

The overall hit rate at K=15 is 2,863/8,581 = 33.4%. The majority of missed keyphrases are bigrams (61.8%), which is expected since bigrams form the largest category of author-assigned keyphrases. Unigrams are missed less frequently (20.2%), likely because single-word terms tend to have higher TF-IDF scores. Longer n-grams (3+) account for 18.1% of misses, often due to exact-match failures where the candidate extraction pipeline produces slight variations of the gold phrase.

3.4 Qualitative Examples

Sample predictions from five randomly selected documents show that both the baseline and full model frequently identify domain-relevant terms that are semantically close to gold keyphrases but do not match after stemming. For instance, for document 1024923 with gold keyphrases including "forward simulations" and "backward simulations", both models correctly retrieve "forward simulation" and "backward simulation" (matching after stemming). However, keyphrases like "refinement mappings" and "prophecy variables" — which are conceptual rather than frequent — are systematically missed.

4 Discussion

4.1 Summary of Findings

Our SAKE framework demonstrates that incorporating document structure into TF-IDF-based keyphrase extraction yields substantial improvements:

1. **Section-aware weighting is the single most impactful feature**, providing ∼29% relative F1 improvement. This validates the core hypothesis that document structure (title, abstract, body) provides strong prior information about keyphrase salience in scientific papers.
2. **Positional weighting provides consistent but modest gains** (∼12% relative F1 improvement), confirming that first-occurrence position correlates with keyphrase importance.
3. **LDA-based semantic relevance does not improve over the structure-aware model.** The 20-topic LDA model's coarse topic assignments appear redundant with the information already captured by position and section features. This is a negative result, but an informative one — it suggests that in the presence of explicit structural signals, lightweight topic modeling adds little value.

4.2 Challenges and Solutions

- **Noisy candidate extraction**: The spaCy POS tagger occasionally produces parsing artifacts. We addressed this with regex-based preprocessing to remove LaTeX, citations, and special tokens before spaCy processing.
- **Hyphenated compounds**: Terms like "graph-based" were being split by spaCy. We solved this by temporarily replacing hyphens with underscores during POS tagging, then restoring them.
- **Exact-match evaluation**: Stemming-based matching misses near-synonyms (e.g., "automaton" vs. "automata"). This is a known limitation of extractive keyphrase evaluation.

4.3 Limitations

- The approach is limited to extractive keyphrases — it cannot generate abstractive keyphrases that do not appear verbatim in the text.

- The LDA model uses a fixed number of topics (20), which may not be optimal. Grid search over topic counts could potentially improve the semantic component.
- Section boundary detection relies on dataset-specific markers (`--T`, `--A`, `--B`). Generalizing to arbitrary documents would require a robust section detector.

4.4 Future Directions

1. **Better semantic representations**: Replace LDA with sentence-level embeddings (e.g., Sentence-BERT) for computing phrase–document similarity.
2. **Learned weight combination**: Instead of fixed multiplicative weights, use a small held-out set to learn optimal combination weights via logistic regression or grid search.
3. **Cross-dataset evaluation**: Test on additional keyphrase benchmarks (SemEval-2010, Inspec, NUS) to assess generalizability.
4. **Abstractive keyphrase generation**: Explore hybrid models that combine extractive candidate ranking with generative keyphrase prediction.

5 Conclusion

We presented SAKE, a simple yet effective unsupervised framework for keyphrase extraction from scientific documents that augments TF-IDF with positional, section-aware, and semantic relevance signals through multiplicative scoring. Our ablation study on the Krapivin benchmark (2,304 documents) reveals a clear hierarchy of feature importance: section-aware weighting contributes the largest gain (+28.7% relative F1@5), followed by positional weighting (+12.0%), while LDA-based semantic relevance provides no additional benefit. The best-performing configuration (TF-IDF + Position + Section with aggressive title/abstract weights) achieves an F1@5 of 0.1668, representing a 56.2% relative improvement over the TF-IDF baseline.

These findings highlight that explicit structural signals in scientific papers — particularly the distinction between title, abstract, and body — are the dominant cue for keyphrase salience in unsupervised settings. Our negative result on LDA suggests that coarse-grained topic models are redundant when structural features are already present, pointing toward richer semantic representations (e.g., contextual embeddings) as the more promising direction for future improvement.

6 Team Responsibilities

- **Donggeun Yoo**: Idea proposing, dataset parsing, text preprocessing, candidate phrase extraction, and implementation of the TF-IDF baseline.
- **Lu Zhang**: Implementation of the positional, section-aware, and semantic relevance modules; conducting ablation experiments; writing the report.

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Declarations

- **Code availability**: All code is available in the companion Jupyter notebooks `ATP.ipynb` and `SAKE_Methodology.ipynb`.
- **Data availability**: The Krapivin dataset is publicly available via HuggingFace (`midas/krapivin`).

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